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Causal WeatherGraph: regime-aware spatio-temporal causal discovery of wind–moisture–cloud pathways from global atmospheric reanalysis data --Manuscript Draft--

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Highlights

Causal WeatherGraph: regime-aware spatio-temporal causal discovery of wind–moisture–cloud pathways from global atmospheric reanalysis data

Xinchen Geng

- Regime-aware spatio-temporal causal discovery for global atmospheric reanalysis
- Physics-constrained candidate graphs reduce the search to 18,018 lagged tests
- Layered validation: FDR control, matched nulls, falsification, adjustment, PCMCI
- Wind–Humidity–Cloud pathway is stable across seven regimes and 1979–2025 splits
- Pathway strongest in midlatitudes; 3,624 chains survive multivariable adjustment

Causal WeatherGraph: regime-aware spatio-temporal causal discovery of wind–moisture–cloud pathways from global atmospheric reanalysis data

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Abstract

Extracting interpretable directed dependency structure from massive, strongly coupled spatio-temporal observations is a central problem in data-driven scientific discovery. We propose Causal WeatherGraph, a regime-aware spatio-temporal causal discovery framework for long-term global atmospheric reanalysis data. The framework aggregates 6-hourly ERA5 fields from 1979 to 2025 into regional multivariate time series, restricts the hypothesis space with a physics-constrained candidate graph, estimates lagged directed dependencies with nested Granger-style regression under seven atmospheric regimes, and screens the resulting 18,018 edge–lag–regime tests with false-discovery control, matched null models, temporal falsification, multivariable adjustment, PCMCI confirmation and stability tests. Of these tests, 14,274 remain significant after false-discovery-rate correction, dominated by Wind $\rightarrow$ Humidity and Humidity $\rightarrow$ Cloud Cover edges. Target-region multivariable adjustment retains 1,920 of 2,218 all-regime discoveries and 3,624 Wind–Humidity–Cloud chains, and PCMCI confirms 50 of 60 top stable core edges. The recovered pathway depends on real temporal alignment, persists across historical and methodological perturbations, and is strongest in midlatitudes. Because median edge-level effects are small and the data are observational, the discovered edges are reported as distributed, physically consistent lagged dependencies rather than interventional causal effects.

Keywords: causal discovery, spatio-temporal data mining, graph-based

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1. Introduction

The global atmosphere is not a simple stack of variables on a regular grid. It is a complex system shaped by dynamical transport, thermodynamic regulation, phase transitions and multiscale circulation. Wind controls the direction and speed of moisture transport, humidity modulates the thermodynamic and microphysical conditions for cloud formation, and cloud cover feeds back on radiative balance, boundary-layer stability and the subsequent evolution of weather systems. For such a system, forecasting future fields is important, but prediction accuracy alone does not answer how variables propagate, which regions are more likely to act as upstream sources, or whether a physical pathway remains stable across seasons, regimes and decades.

Data-driven atmospheric modelling has made rapid progress on several fronts. Global neural forecasting systems such as FourCastNet, Pangu-Weather and GraphCast now match or surpass operational numerical weather prediction at medium range, and hybrid dynamical and probabilistic ensemble successors extend this skill further [1–5]. In parallel, spatio-temporal graph learning has moved from fixed adjacency matrices toward adaptive and hierarchical graph structures learned from data, including multi-level graphs tailored to weather variables and stations [6–8]. On the interpretability side, causal discovery for Earth-system time series combines temporal precedence, conditional-independence testing and domain knowledge to recover directed lagged dependencies among climate variables [9–11]. Together, these strands supply mature tools for forecasting, for learning spatial structure and for testing directed dependence in multivariate time series.

Despite this progress, none of these strands directly answers how a physical pathway propagates through the atmosphere. The internal graph or attention weights of a forecasting model are tuned for predictive skill and need not correspond to directed physical pathways, while most causal-discovery methods are validated on systems far smaller and cleaner than global reanalysis, in which seasonal cycles, spatial smoothing, spatial autocorrelation, common weather systems and bidirectional coupling can all create apparently directional statistical relationships. We therefore ask the following question: can long-term global atmospheric reanalysis data be used to construct an

interpretable regional lagged dependency graph that identifies stable, reproducible and dynamically plausible temporal propagation among wind, humidity and cloud cover? We focus on the Wind $\rightarrow$ Humidity $\rightarrow$ Cloud Cover pathway. This pathway has a clear physical interpretation: wind organizes moisture transport, humidity anomalies then modify the conditions for cloud formation, and the response is observed as a change in cloud cover. Precisely because of the dependence structure of observational data, this picture must be tested rather than assumed.

To address this problem, we propose Causal WeatherGraph: Regime-Aware Spatio-Temporal Causal Discovery for Global Atmospheric Dynamics. The framework converts gridded atmospheric fields into regional multivariate time series, restricts the search space using physically motivated candidate edges, and applies Granger-style nested regression tests to identify lagged incremental predictive relationships. The term "causal" is used in a causal-discovery-inspired sense. It denotes a lagged directed dependency in which a source lag still provides significant incremental explanatory power for the current target state after controlling for the target history. It does not denote an interventional causal effect. From an information-processing standpoint, the task is one of scalable and statistically controlled structure discovery: the raw archive contains more than 5.6×10^8 grid-level observations, and an unconstrained edge search over region-variable nodes would be both computationally prohibitive and statistically fragile under strong spatio-temporal dependence. Figure 1 provides an overview of the complete framework and its headline results.

We make three main contributions:

- **Scalable physics-constrained graph construction.** We compress high-dimensional gridded fields into interpretable region-variable nodes and restrict the edge hypothesis space to physically admissible candidates, reducing an intractable unconstrained search to 18,018 regime-conditioned lagged tests while preserving physical interpretability and computational feasibility.
- **Regime-aware structure estimation.** We estimate separate graph structures for the all-sample, seasonal, high-humidity, normal-humidity, high-cloud and normal-cloud regimes rather than forcing all samples into a single static graph, making state-dependent dependency structure an explicit object of analysis.

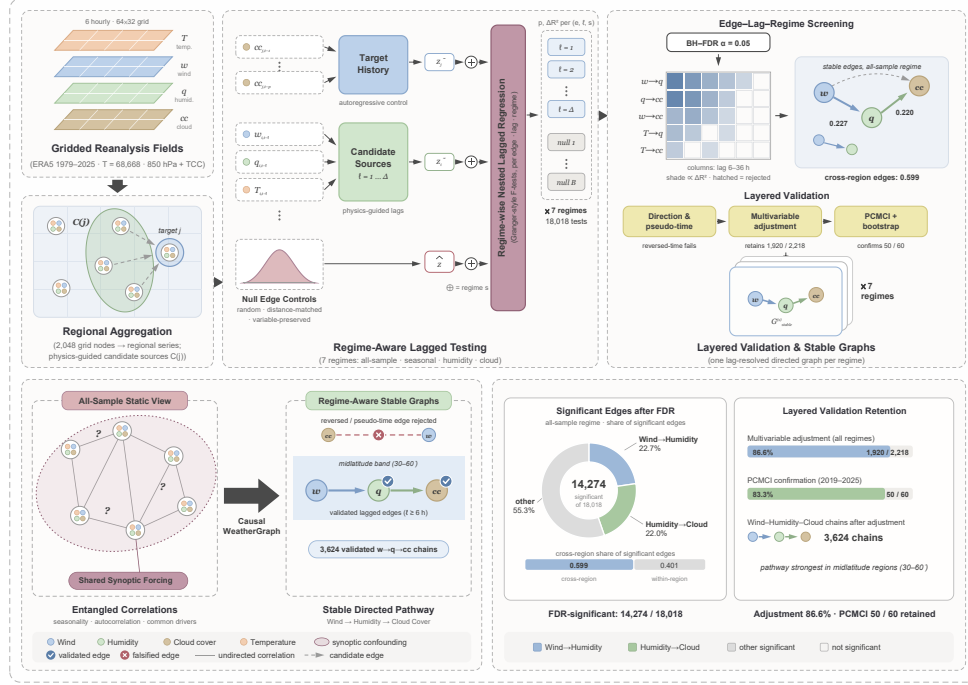


Figure 1: Overview of the Causal WeatherGraph framework and headline results. ERA5 fields (1979–2025) are aggregated into regional multivariate series, constrained by physics-guided candidate edges, and screened by regime-wise nested lagged regressions with BH-FDR control and a layered validation chain; 14,274 of 18,018 (edge, lag, regime) tests remain significant, dominated by Wind $\rightarrow$ Humidity and Humidity $\rightarrow$ Cloud Cover.

- Layered statistical validation.** We couple the discovery step with a reusable validation chain — random, distance-matched and variable-preserved edge controls; directionality and pseudo-time falsification; target-region multivariable adjustment; effect-size and confidence-interval reporting; PCMCi confirmation; block bootstrap; long-lag and humidity-mediation tests; latitude-band stratification; matched-sample regime analysis; candidate- k expansion; historical-period stability; and preprocessing and region-aggregation robustness — so that the reported structure is explicitly controlled for the dependence artefacts that observational spatio-temporal data induce.

2. Related Work

Figure 2 situates Causal WeatherGraph within four strands of related research: data-driven atmospheric prediction, spatio-temporal graph and multivariate time-series learning, causal discovery for Earth-system time series, and regime-dependent atmospheric interactions.

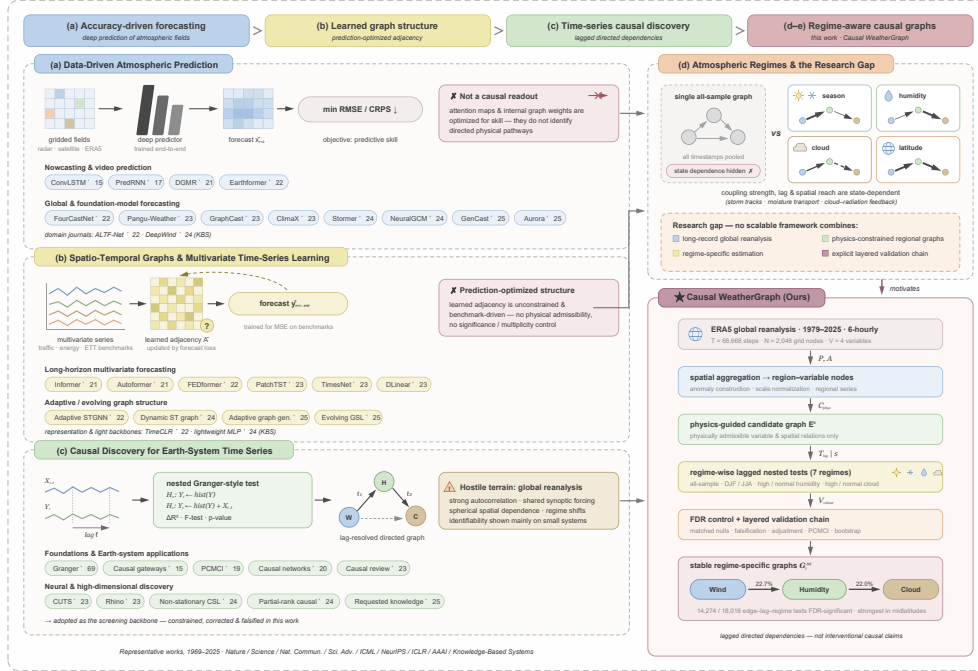


Figure 2: Positioning of Causal WeatherGraph within related work. Predictive models (a–b) optimize forecast skill rather than directed physical pathways, causal discovery (c) is strained by the dependence structure of global reanalysis, and single all-sample graphs (d) obscure regime differences; Causal WeatherGraph (e) closes this gap with physics-guided candidates, regime-wise lagged tests and layered validation.

2.1. Data-driven atmospheric prediction

Deep spatio-temporal prediction established that neural networks can learn nonlinear evolution from radar, satellite and reanalysis fields. ConvLSTM and PredRNN introduced convolutional recurrent memory for precipitation nowcasting and predictive learning [12, 13]; more recent systems use generative modelling and space–time attention to improve probabilistic

nowcasting and long-range dependence representation [14, 15]. Recent work has also moved toward transferable atmospheric representations: ClimaX formulates weather and climate modelling as a foundation-model problem [16], while Stormer studies the scaling of transformer weather models [17]. In the applied forecasting literature, ALTF-Net applies attention to long-horizon air-temperature forecasting, while DeepWind combines heterogeneous spatial and temporal information for wind forecasting [18, 19]. These studies demonstrate the value of learned multiscale representations for atmospheric series, but their primary objective is predictive accuracy rather than the identification and falsification of interpretable directed physical pathways.

Global data-driven forecasting has advanced from Fourier neural operators and three-dimensional neural architectures to graph-based, hybrid dynamical and probabilistic ensemble systems. FourCastNet, Pangu-Weather and GraphCast established competitive medium-range global forecasting [1–3]; WeatherBench 2 subsequently provided a standardized evaluation framework [20]. NeuralGCM couples machine-learned components with a differentiable atmospheric dynamical core, GenCast extends global prediction to probabilistic ensembles, and Aurora demonstrates a foundation model spanning multiple Earth-system forecasting tasks [4, 5, 21]. Collectively, this recent literature shows that atmospheric states are naturally modelled through multiscale spatial interactions. Nevertheless, the internal graph or attention weights of a forecasting model are optimized for prediction and should not automatically be interpreted as causal or physically directed edges.

2.2. Spatio-temporal graphs and multivariate time-series learning

Long-horizon multivariate forecasting has evolved rapidly. Informer introduced efficient attention for long sequences, Autoformer incorporated decomposition and autocorrelation, and FEDformer used frequency-enhanced decomposition [22–24]. PatchTST and TimesNet subsequently developed patch-based channel modelling and multi-period two-dimensional temporal representations, while DLinear showed that carefully designed linear baselines remain highly competitive [25–27]. This body of work motivates careful separation between representational complexity and scientifically interpretable dependence.

Recent spatio-temporal graph research increasingly learns or updates graph topology instead of assuming a fixed adjacency matrix. Adaptive graph neural networks, dynamic spatial-aware graph transformers and unified adaptive graph-generation methods have improved forecasting where spatial

relations change over time [7, 28, 29]. Evolving graph structure learning further links changing multivariate dependencies to forecasting performance [8]. In weather forecasting specifically, HiSTGNN learns a hierarchical graph that couples a station-level global graph with variable-level local graphs and passes information bidirectionally between the two levels, improving multi-station, multi-variable prediction [6]. Complementary work develops self-supervised time-series representations and lightweight multivariate forecasting backbones [30, 31]. Although most of these methods were evaluated on traffic, industrial or generic benchmark series, they provide relevant computational ideas for large spatio-temporal systems. The distinction is that Causal WeatherGraph does not learn an unconstrained predictive adjacency: it begins from physically admissible variable and spatial relations, estimates explicit positive-lag tests, applies multiplicity control, and reports graph edges only within a layered falsification framework.

2.3. Causal discovery for Earth-system time series

Granger causality formalized temporal precedence and incremental prediction for lagged time series [9]. Earth-system causal discovery subsequently combined conditional-independence testing, graphical models and domain constraints to identify gateways, mediators and model-process relationships [10, 32, 33]. A recent cross-disciplinary review emphasizes that causal discovery from observational dynamical systems requires explicit assumptions about temporal order, hidden confounding, non-stationarity and functional form [11]. CUTS extended neural causal discovery to irregularly sampled time series, while Rhino modelled temporal causal relations with history-dependent noise [34, 35]. Recent studies additionally address high-dimensional non-stationary causal structure learning, partial-rank-correlation causal networks and the use of dynamically requested background knowledge [36–38]. These developments are closely related to the present objective, but global atmospheric reanalysis introduces unusually strong autocorrelation, shared synoptic forcing, spherical spatial dependence and regime shifts. We therefore use scalable Granger-style nested regression as the screening backbone and constrain its interpretation through physics-guided candidates, false-discovery control, temporal falsification, multivariable adjustment, PCMCI confirmation and robustness analysis.

2.4. Atmospheric regimes and the research gap

Atmospheric interactions are state dependent. Season, humidity background, cloud regime and latitude-band distribution can alter coupling strength, lag and spatial reach. Synoptic-scale circulation organizes moisture transport in midlatitude cyclones, while storm tracks interact with water vapour, latent heating, clouds and radiation [39–41]. Consequently, a single graph estimated from all timestamps may obscure physically meaningful changes in pathway prevalence or strength. Existing forecasting studies usually assess skill by lead time and variable, whereas causal-discovery studies often prioritize methodological identifiability on smaller systems. The unresolved gap is a scalable framework that combines long-record global reanalysis, physically constrained regional graphs, regime-specific estimation and an explicit validation chain. Causal WeatherGraph addresses this gap by comparing pathway counts, edge-type fractions, cross-region fractions, lags, chain scores and stability across seasonal, humidity and cloud regimes.

3. Methodology

3.1. Problem formulation and framework overview

Let

$$\mathcal{X}^{\text{raw}} = \{x_{t,n,v}^{\text{raw}}\} \in \mathbb{R}^{T \times N \times V} \quad (1)$$

denote a multivariate atmospheric field observed at time index $t \in \{1, \dots, T\}$, grid node $n \in \{1, \dots, N\}$ and variable $v \in \mathcal{V}$. In this study, $T = 68,668$, $N = 2,048$ and

$$\mathcal{V} = \{\text{temperature, humidity, wind, cloud}\}, \quad V = 4. \quad (2)$$

The observations span 1979-01-01 00:00:00 to 2025-12-31 18:00:00 at a 6-hour sampling interval. Wind is represented by scalar wind speed, obtained from the available speed field or from the Euclidean magnitude of zonal and meridional components, $w = (u^2 + v^2)^{1/2}$.

Our objective is to estimate a family of regime-specific, lag-resolved directed graphs

$$\{\mathcal{G}_\ell^{(s)} = (\mathcal{U}, \mathcal{E}_\ell^{(s)}, \mathcal{W}_\ell^{(s)})\}_{s \in \mathcal{S}, \ell=1}^L, \quad (3)$$

where $\mathcal{U} = \mathcal{R} \times \mathcal{V}$ is the set of region-variable nodes, $\mathcal{R}$ is the set of spatial regions, ℓ is a positive lag, and s indexes an atmospheric regime. An edge $(i, a) \rightarrow (j, b)$ means that variable a in source region i provides statistically

significant incremental information about variable b in target region j after the target history has been controlled. Thus, the estimand is a conditional lagged directed dependency rather than an interventional causal effect.

The complete pipeline can be summarized as

$$\mathcal{X}^{\text{raw}} \xrightarrow{\mathcal{P}} \mathcal{Z} \xrightarrow{\mathcal{A}} \mathbf{X} \xrightarrow{\mathcal{C}_{\text{phys}}} \mathcal{E}^0 \xrightarrow{\mathcal{T}_{\text{lag}} | s} \mathcal{G}^{(s)} \xrightarrow{\mathcal{V}_{\text{robust}}} \mathcal{G}_{\text{stable}}^{(s)}, \quad (4)$$

where $\mathcal{P}$ denotes temporal preprocessing, $\mathcal{A}$ spatial aggregation, $\mathcal{C}_{\text{phys}}$ physics-guided candidate generation, $\mathcal{T}_{\text{lag}}$ lagged nested-model testing and $\mathcal{V}_{\text{robust}}$ the collection of stability and falsification procedures.

3.2. Data source, variables and temporal harmonization

The analysis uses the ERA5 global atmospheric reanalysis [42]. The 1979-01-01 to 2018-12-31 segment and the 2019-01-01 to 2023-01-10 segment were obtained from WeatherBench 2 ERA5 products [20]. The continuation from 2023-01-11 to 2025-12-31 was downloaded from the Copernicus Climate Data Store and harmonized to the same 6-hourly, 64×32 regular latitude–longitude grid. Overlapping timestamps were de-duplicated and all variables were intersected on a common time index. The resulting sequence contains 68,668 timestamps from 1979-01-01 00:00 UTC through 2025-12-31 18:00 UTC, with no missing or duplicated 6-hourly timestamps in the aligned analysis array.

Temperature, specific humidity, zonal wind and meridional wind are instantaneous ERA5 fields at 850 hPa. Their native units are K, kg kg^{-1} and m s^{-1} , respectively. Scalar wind speed is computed as $w = (u^2 + v^2)^{1/2}$. Total cloud cover is a dimensionless single-level field in the range 0–1 and represents the fraction of the grid box covered by cloud integrated across model levels. Thus, the pathway combines lower-tropospheric dynamical and moisture fields at 850 hPa with a column-integrated cloud-cover response. All fields are subsequently transformed to standardized anomalies, so reported regression coefficients are dimensionless standardized effects.

3.3. Anomaly construction and scale normalization

Short missing intervals were linearly interpolated for at most six time steps; remaining endpoints were forward- and backward-filled. Grid nodes for which more than 20% of node–variable entries were missing were excluded before subsequent analysis. To suppress the dominant annual cycle, we removed a calendar-month climatology independently for every grid node and

variable:

$$a_{t,n,v} = x_{t,n,v}^{\text{raw}} - \mu_{m(t),n,v}, \quad \mu_{m,n,v} = \frac{1}{|\mathcal{T}_m|} \sum_{t' \in \mathcal{T}_m} x_{t',n,v}^{\text{raw}}, \quad (5)$$

where $m(t) \in \{1, \dots, 12\}$ is the month containing time t and $\mathcal{T}_m = \{t : m(t) = m\}$. Each anomaly series was then standardized over time:

$$z_{t,n,v} = \frac{a_{t,n,v} - \bar{a}_{n,v}}{\max(\sigma_{n,v}, 10^{-8})}, \quad \bar{a}_{n,v} = \frac{1}{T} \sum_{t=1}^T a_{t,n,v}, \quad \sigma_{n,v}^2 = \frac{1}{T} \sum_{t=1}^T (a_{t,n,v} - \bar{a}_{n,v})^2. \quad (6)$$

This transformation makes regression coefficients comparable across variables and prevents high-variance fields from dominating the edge ranking. The main analysis did not apply first differencing. Robustness experiments additionally considered monthly anomalies with first differences, first differences without monthly climatology removal, centered rolling anomalies, and standardization after rather than before regional aggregation.

3.4. Spherical regionalization

Direct causal screening over all $N \times V$ grid-variable nodes is both computationally expensive and difficult to interpret. We therefore partitioned the grid into a set of regions $\mathcal{R} = \{\mathcal{R}_1, \dots, \mathcal{R}_R\}$. In the principal setting, the requested number of regions was $R_0 = 64$. The implementation uses

$$n_\phi = \text{round}\left(\sqrt{R_0/2}\right), \quad n_\lambda = \lceil R_0/n_\phi \rceil, \quad (7)$$

equal-width latitude and longitude bins. This gives $n_\phi = 6$, $n_\lambda = 11$ and hence $R = 66$ non-empty regional nodes. The regional time series is the arithmetic mean of standardized grid-level signals,

$$X_{t,r,v} = \frac{1}{|\mathcal{R}_r|} \sum_{n \in \mathcal{R}_r} z_{t,n,v}, \quad \mathbf{X} \in \mathbb{R}^{T \times R \times V}. \quad (8)$$

For mapping and neighbourhood construction, the regional latitude is the arithmetic mean of member latitudes, whereas longitude is computed by a circular mean:

$$\bar{\lambda}_r = \text{atan2} \left(\frac{1}{|\mathcal{R}_r|} \sum_{n \in \mathcal{R}_r} \sin \lambda_n, \frac{1}{|\mathcal{R}_r|} \sum_{n \in \mathcal{R}_r} \cos \lambda_n \right). \quad (9)$$

Alternative regionalizations with 32, 64 and 128 target bins and spherical k -means partitions were evaluated to test sensitivity to spatial resolution and partition geometry.

Table 1: Data and main experimental configuration.

Item	Setting
Time range	1979-01-01 00:00:00 to 2025-12-31 18:00:00
Temporal resolution	6 hours
Time steps	68,668
Raw array shape	(68668, 2048, 4)
Regional array shape	(68668, 66, 4)
Variables	temperature, humidity, wind, cloud_cover
Physical fields	850-hPa temperature, specific humidity and wind speed; total cloud cover
Native units	K; kg kg ⁻¹ ; m s ⁻¹ ; cloud fraction 0–1
Data products	WeatherBench 2 ERA5 through 2023-01-10; CDS ERA5 thereafter
Spatial grid	64 × 32 regular latitude–longitude grid (2,048 nodes)
Region aggregation	latlon_bins, target 64, actual 66
Main method	Granger-style OLS
Maximum lag	max_lag = 3, corresponding to 6–18 hours
Candidate neighbourhood	candidate_k_nearest = 2
Multiple testing control	Benjamini-Hochberg FDR, q_value < 0.05

3.5. Physics-constrained candidate graph

An unconstrained directed search would require $O(R^2 V^2 L |\mathcal{S}|)$ tests and would devote most of the multiple-testing budget to physically uninformative variable pairs. We instead define a sparse candidate graph $\mathcal{G}^0 = (\mathcal{U}, \mathcal{E}^0)$ from two relation sets:

$$\mathcal{P}_{\text{local}} = \{W \rightarrow H, H \rightarrow C, T \rightarrow H, T \rightarrow C, W \rightarrow C\}, \quad (10)$$

$$\mathcal{P}_{\text{spatial}} = \{W \rightarrow H, H \rightarrow H, H \rightarrow C, W \rightarrow C\}, \quad (11)$$

where T , H , W and C denote temperature, humidity, wind and cloud cover. Local relations connect variables within the same region. Spatial relations connect each target region j to its k nearest possible source regions.

Nearest neighbours are determined by great-circle distance. For regional coordinates (ϕ_i, λ_i) and (ϕ_j, λ_j) in radians,

$$a_{ij} = \sin^2\left(\frac{\phi_i - \phi_j}{2}\right) + \cos \phi_i \cos \phi_j \sin^2\left(\frac{\lambda_i - \lambda_j}{2}\right), \quad (12)$$

$$d_{ij} = 2R_{\oplus} \arcsin(\sqrt{a_{ij}}), \quad R_{\oplus} = 6,371 \text{ km}. \quad (13)$$

Let $\mathcal{N}_k(j)$ denote the k nearest regions to target j . Then

$$\mathcal{E}^0 = \{(r, a) \rightarrow (r, b) : (a, b) \in \mathcal{P}_{\text{local}}\} \cup \{(i, a) \rightarrow (j, b) : i \in \mathcal{N}_k(j), (a, b) \in \mathcal{P}_{\text{spatial}}\}. \quad (14)$$

With $R = 66$ and $k = 2$, this construction yields $66 \times 5 + 66 \times 2 \times 4 = 858$ directed candidates. Testing three lags therefore produces 2,574 edge-lag hypotheses per regime, rather than millions of all-to-all hypotheses. We repeated the analysis with $k \in \{4, 6, 8\}$ to measure dependence on candidate-neighbourhood size.

3.6. Regime-conditioned sampling

Atmospheric coupling is non-stationary, so a single graph may average over dynamically distinct states. For every regime $s \in \mathcal{S}$, define a binary mask $m_t^{(s)} \in \{0, 1\}$ and the corresponding sample set

$$\mathcal{I}_s = \{t \in \{L + 1, \dots, T\} : m_t^{(s)} = 1\}. \quad (15)$$

The seasonal masks select December–February (DJF) and June–August (JJA). For humidity and cloud-cover states, we first form spatially averaged regional indices

$$g_t^H = \frac{1}{R} \sum_{r=1}^R X_{t,r,H}, \quad g_t^C = \frac{1}{R} \sum_{r=1}^R X_{t,r,C}. \quad (16)$$

Writing $Q_p(g)$ for the empirical p -quantile of a series g , the masks are

$$m_t^{(\text{highH})} = \mathbb{I}[g_t^H \geq Q_{0.8}(g^H)], \quad m_t^{(\text{normalH})} = \mathbb{I}[Q_{0.2}(g^H) < g_t^H < Q_{0.8}(g^H)], \quad (17)$$

$$m_t^{(\text{highC})} = \mathbb{I}[g_t^C \geq Q_{0.8}(g^C)], \quad m_t^{(\text{normalC})} = \mathbb{I}[g_t^C < Q_{0.8}(g^C)]. \quad (18)$$

Together with the all-sample mask, this gives

$$\mathcal{S} = \{\text{all}, \text{DJF}, \text{JJA}, \text{highH}, \text{normalH}, \text{highC}, \text{normalC}\}. \quad (19)$$

The regime mask is applied to the response time t ; lagged predictors retain their physically preceding observations.

3.7. Conditional lagged-dependency estimator

Consider a candidate $e : (i, a) \rightarrow (j, b) \in \mathcal{E}^0$. Let

$$y_t = X_{t,j,b}, \quad x_t^{(e,\ell)} = X_{t-\ell,i,a}, \quad \mathbf{c}_{t,j,b} = [X_{t-1,j,b}, \dots, X_{t-L,j,b}]^\top. \quad (20)$$

The principal experiment uses $L = 3$, corresponding to 6, 12 and 18 hours, and controls the three own lags of the target series. For each (e, ℓ, s) , we fit the restricted and full models on $t \in \mathcal{I}_s$:

$$\mathcal{M}_R : \quad y_t = \alpha_R + \boldsymbol{\gamma}^\top \mathbf{c}_{t,j,b} + \varepsilon_t^R, \quad (21)$$

$$\mathcal{M}_F : \quad y_t = \alpha_F + \boldsymbol{\gamma}^\top \mathbf{c}_{t,j,b} + \beta_{e,\ell}^{(s)} x_t^{(e,\ell)} + \varepsilon_t^F. \quad (22)$$

Parameters are estimated by ordinary least squares,

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \sum_{t \in \mathcal{I}_s} \left(y_t - \mathbf{d}_t^\top \boldsymbol{\theta} \right)^2, \quad (23)$$

where $\mathbf{d}_t$ is the relevant restricted or full design vector. The null hypothesis is

$$H_0^{(e,\ell,s)} : \beta_{e,\ell}^{(s)} = 0, \quad (24)$$

meaning that the source lag does not improve prediction after target persistence has been removed. With residual sums of squares RSS_R and RSS_F , the one-degree-of-freedom nested test is

$$F_{e,\ell}^{(s)} = \frac{(\text{RSS}_R - \text{RSS}_F) / 1}{\text{RSS}_F / (N_s - L - 2)}, \quad N_s = |\mathcal{I}_s|. \quad (25)$$

The corresponding upper-tail $F_{1, N_s - L - 2}$ probability gives the raw p -value. We retain the signed standardized coefficient $\hat{\beta}_{e,\ell}^{(s)}$ as the effect estimate.

To distinguish statistical detectability from practical magnitude, we also report

$$\Delta R^2 = \frac{\text{RSS}_R - \text{RSS}_F}{\text{TSS}}, \quad R_{\text{partial}}^2 = \frac{\text{RSS}_R - \text{RSS}_F}{\text{RSS}_R}, \quad (26)$$

and Cohen's $f^2 = (\text{RSS}_R - \text{RSS}_F) / \text{RSS}_F$. Model-based coefficient standard errors and 95% intervals are obtained from the full OLS model. Because model-based intervals do not by themselves address serial dependence, the ten strongest stable edges in each core pathway are also evaluated using year-block bootstrap intervals across the 47 calendar years.

For repeated robustness experiments, we use an algebraically equivalent Frisch–Waugh–Lovell residualization. If $\mathbf{C}$ is the target-history design matrix including the intercept and $\mathbf{M}_C = \mathbf{I} - \mathbf{C}(\mathbf{C}^\top \mathbf{C})^{-1} \mathbf{C}^\top$, then

$$\tilde{\mathbf{y}} = \mathbf{M}_C \mathbf{y}, \quad \tilde{\mathbf{x}} = \mathbf{M}_C \mathbf{x}, \quad \hat{\beta} = \frac{\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}}}{\tilde{\mathbf{x}}^\top \tilde{\mathbf{x}}}. \quad (27)$$

The restricted target fit is reused for all candidate sources associated with the same target, reducing repeated computation without changing the tested linear model.

Target-region multivariable adjustment. The principal model controls the target variable's own three lags. As a stronger confounding sensitivity analysis, we repeat the complete graph estimation while controlling all three lags of temperature, humidity, wind and cloud cover in the target region. For an intra-region candidate, the exact source-lag term being tested is excluded from the restricted control matrix. This model asks whether a source retains incremental information beyond the recent multivariate atmospheric state of the target region.

3.8. False-discovery control and graph assembly

Because each regime contains many edge-lag hypotheses, raw significance is insufficient. Tests are partitioned by regime s and target variable b , matching the implementation. Benjamini–Hochberg false-discovery control [43] is applied within each group. Within a group containing M hypotheses, let $p_{(1)} \leq \dots \leq p_{(M)}$ be ordered raw p -values. The Benjamini–Hochberg adjusted value is

$$q_{(i)} = \min \left\{ 1, \min_{j \geq i} \frac{M}{j} p_{(j)} \right\}. \quad (28)$$

An edge-lag hypothesis is selected when $q_{e,\ell}^{(s)} < \alpha$ with $\alpha = 0.05$. Its signed graph weight is

$$\omega_{e,\ell}^{(s)} = \text{sign}(\hat{\beta}_{e,\ell}^{(s)}) |\hat{\beta}_{e,\ell}^{(s)}| \left[-\log_{10}(q_{e,\ell}^{(s)} + 10^{-12}) \right]. \quad (29)$$

Thus, ω jointly reflects effect magnitude, sign and multiplicity-adjusted evidence. The lag-specific weighted adjacency tensor is

$$A_{(i,a),(j,b),\ell}^{(s)} = \omega_{e,\ell}^{(s)} \mathbb{I}[e : (i, a) \rightarrow (j, b) \in \mathcal{E}^0, q_{e,\ell}^{(s)} < 0.05]. \quad (30)$$

Multiple significant lags for the same region–variable pair are retained because they represent distinct tested time scales.

3.9. Compositional Wind–Humidity–Cloud pathways

The proposed graph is evaluated not only edge by edge, but also through physically interpretable two-step paths. In regime s , the Wind–Humidity–Cloud path set is

$$\mathcal{P}_{\text{WHC}}^{(s)} = \left\{ (i, h, j, \ell_1, \ell_2) : A_{(i,W),(h,H),\ell_1}^{(s)} \neq 0, \right. \\ \left. A_{(h,H),(j,C),\ell_2}^{(s)} \neq 0 \right\}. \quad (31)$$

The matching condition on region h makes humidity a genuine graph intermediate rather than merely the same variable label. For a path $p = (i, h, j, \ell_1, \ell_2)$, we define

$$\tau_p = \ell_1 + \ell_2, \quad \Omega_p = \left| \omega_{(i,W) \rightarrow (h,H), \ell_1}^{(s)} \right| + \left| \omega_{(h,H) \rightarrow (j,C), \ell_2}^{(s)} \right|. \quad (32)$$

Reported pathway statistics are

$$N_{\text{WHC}}^{(s)} = \left| \mathcal{P}_{\text{WHC}}^{(s)} \right|, \quad \bar{\tau}^{(s)} = \frac{1}{N_{\text{WHC}}^{(s)}} \sum_p \tau_p, \quad \bar{\Omega}^{(s)} = \frac{1}{N_{\text{WHC}}^{(s)}} \sum_p \Omega_p, \quad (33)$$

together with the number of unique intermediate humidity regions. Structural differences between two regime graphs are summarized by Jaccard distance,

$$D_J(s_1, s_2) = 1 - \frac{\left| \mathcal{E}^{(s_1)} \cap \mathcal{E}^{(s_2)} \right|}{\left| \mathcal{E}^{(s_1)} \cup \mathcal{E}^{(s_2)} \right|}, \quad (34)$$

where regime comparison uses unique directed region–variable edges after collapsing over lag.

3.10. Stability selection, falsification and mechanism checks

Temporal block bootstrap. To preserve short-range serial dependence, we sample contiguous blocks of length B with replacement from the valid indices of each regime until a bootstrap series of size N_s is obtained [44]. For B_{boot} replicates, edge stability is

$$\hat{\pi}_{e,\ell}^{(s)} = \frac{1}{B_{\text{boot}}} \sum_{b=1}^{B_{\text{boot}}} \mathbb{I} \left[q_{e,\ell}^{(s,b)} < 0.05 \right]. \quad (35)$$

The default stability analysis uses 20 replicates and defines a stable edge as one satisfying $q < 0.05$, $\hat{\pi} \geq 0.6$, and $|\hat{\beta}|$ at or above the median absolute coefficient among significant edges.

Graph null models. We compare the physical candidate graph against three graph-level nulls: (i) fully random source region, target region and variable pairs; (ii) variable-preserved randomization, which keeps (a, b) fixed but randomizes (i, j) ; and (iii) distance-matched randomization, which preserves (a, b) and approximately matches d_{ij} . For any pathway type p , its enrichment relative to a null graph is

$$\text{ER}_p = \frac{N_p^{\text{physical}} / N_{\text{tested}}^{\text{physical}}}{N_p^{\text{null}} / N_{\text{tested}}^{\text{null}}}. \quad (36)$$

Temporal and directional falsification.. We recompute the tests after reversing candidate directions, randomly permuting the time axis, reversing the complete temporal order (future-to-past), and circularly shifting the relevant source variable by 1,460 six-hour steps, equivalent to one year. A physically meaningful lagged dependency should weaken when temporal alignment or direction is destroyed.

Humidity mediation diagnostic.. For every significant Wind $\rightarrow$ Cloud edge, we first estimate a direct model containing cloud own lags and the tested wind lag, producing $\hat{\beta}_{W \rightarrow C}^{\text{direct}}$. We then add all target region humidity lags $H_{t-1,j}, \dots, H_{t-L,j}$ and obtain $\hat{\beta}_{W \rightarrow C}^{\text{ctrlH}}$. The diagnostic attenuation is

$$\rho = 1 - \frac{|\hat{\beta}_{W \rightarrow C}^{\text{ctrlH}}|}{|\hat{\beta}_{W \rightarrow C}^{\text{direct}}|}. \quad (37)$$

A positive ρ is consistent with partial transmission through humidity, although it is not a nonparametrically identified natural indirect effect.

Sensitivity analyses.. The maximum lag is extended from $L = 3$ to $L = 12$ (72 hours) for the principal pathways. Spatial heterogeneity is assessed over tropical ($|\phi| < 23.5^\circ$), midlatitude ($23.5^\circ \leq |\phi| < 60^\circ$) and polar ($|\phi| \geq 60^\circ$) target regions. We additionally vary preprocessing, regionalization, candidate-neighbourhood size and historical period. Selected high-stability edges are reassessed using PCMCI with partial-correlation conditional-independence tests. PCMCI confirmation uses the 2019–2025 segment at the original 6-hour resolution, the top 20 all-regime stable edges from each core edge type, $\tau_{\max} = 3$, a maximum PC conditioning-set dimension of three and Benjamini–Hochberg-adjusted link probabilities.

To check whether high-state versus normal-state graph differences are caused only by unequal sample sizes, normal-humidity and normal-cloud timestamps are randomly subsampled to the corresponding high-regime sample size in 20 repetitions. Graph Jaccard distance and edge-type counts are recomputed for every matched sample. Across all analyses, the term “causal” is restricted to causal-discovery- inspired lagged dependence under the stated controls; hidden common drivers, measurement error and contemporaneous feedback remain possible.

4. Experimental Results

4.1. Overall detection of regional lagged directed dependencies

The main experiment tested 18,018 edge-lag-regime combinations, of which 14,274 were significant after FDR correction, giving an overall significant ratio of 0.792. In the all regime, 2,218 of 2,574 edge-lag combinations were significant. Wind $\rightarrow$ Humidity accounted for 0.227 of significant edges, Humidity $\rightarrow$ Cloud Cover accounted for 0.220, and cross-region significant edges accounted for 0.599. The mean lag of the two core edge types was close to two 6-hour time steps, approximately 12 hours.

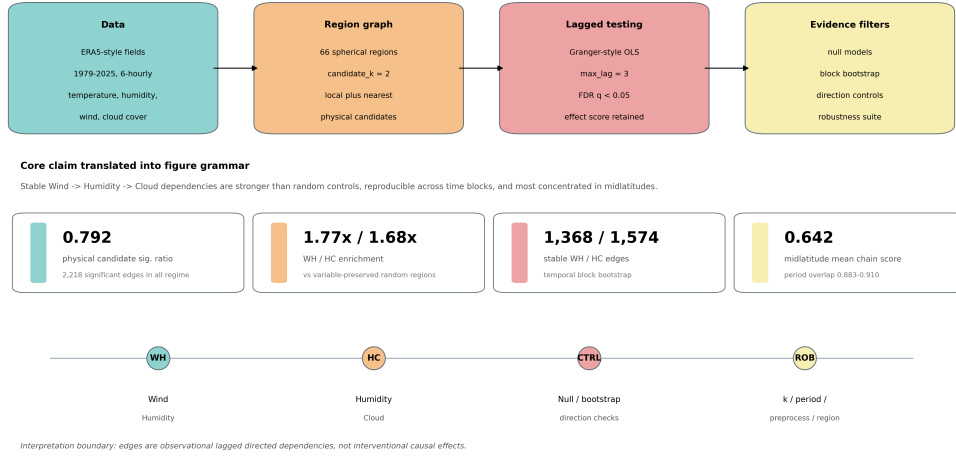


Figure 3: Evidence stack of the main experiments: data scale, candidate graph, pathway statistics and robustness evidence for Wind $\rightarrow$ Humidity $\rightarrow$ Cloud Cover.

Table 2: Main experimental results across regimes.

Regime	Tested edges	Significant edges	Wind $\rightarrow$ Humidity fraction	Humidity $\rightarrow$ Cloud fraction	Cross-region fraction	Mean WH lag	Mean HC lag
all	2,574	2,218	0.227	0.220	0.599	2.006	1.977
DJF	2,574	2,017	0.243	0.213	0.500	1.992	1.979
JJA	2,574	2,023	0.218	0.218	0.589	1.975	1.989
high_humidity	2,574	1,842	0.224	0.204	0.571	1.990	1.952
normal_humidity	2,574	2,140	0.228	0.220	0.590	2.008	1.979
high_cloud	2,574	1,843	0.231	0.213	0.571	1.988	1.969
normal_cloud	2,574	2,191	0.227	0.220	0.595	2.000	1.969

4.2. Spatial organization of the Wind–Humidity–Cloud Cover pathway

Pathway-level statistics showed that Wind $\rightarrow$ Humidity $\rightarrow$ Cloud Cover was not a structure assembled from a small number of local edges. In the

all regime, 3,746 WHC chains were identified, covering all 66 intermediate humidity regions. The mean total lag was 3.983 time steps, approximately 24 hours. DJF contained 3,215 chains with a mean chain score of 0.578, whereas JJA contained 2,998 chains with a mean chain score of 0.553.

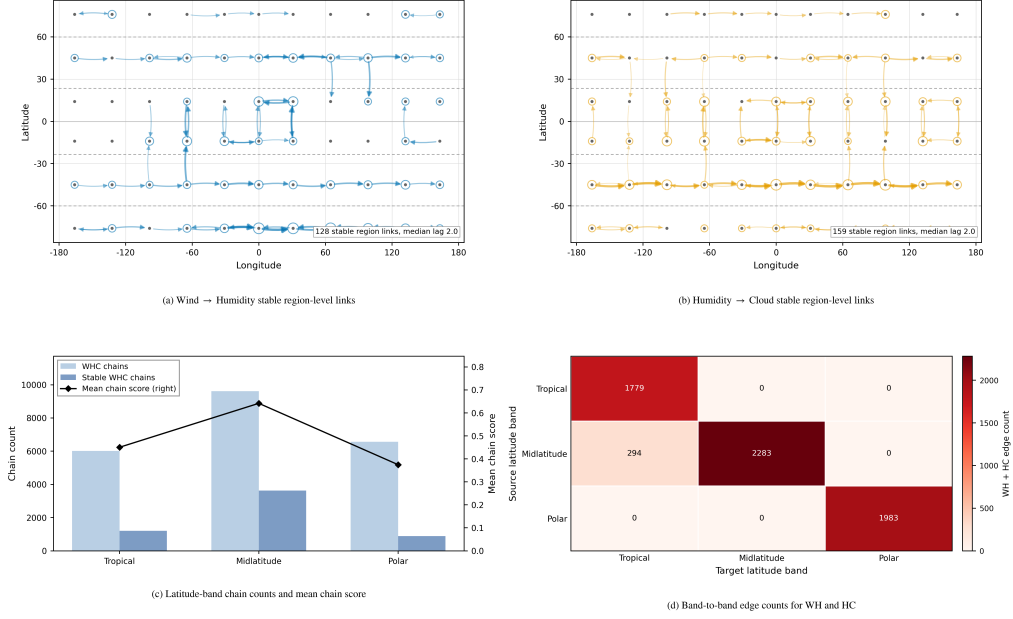


Figure 4: Spatial structure of the Wind $\rightarrow$ Humidity and Humidity $\rightarrow$ Cloud Cover edges, separating local from cross-region dependencies.

Table 3: Wind $\rightarrow$ Humidity $\rightarrow$ Cloud Cover chain statistics.

Regime	WHC chains	Intermediate humidity regions	Mean total lag	Mean chain score
all	3,746	66	3.983	0.468
DJF	3,215	66	3.976	0.578
JJA	2,998	66	3.963	0.553
high_humidity	2,442	62	3.939	0.551
normal_humidity	3,522	65	3.988	0.486
high_cloud	2,612	65	3.962	0.499
normal_cloud	3,667	66	3.972	0.468

4.3. Regime-aware analysis of state-dependent structure

Regime comparison showed that the core pathway persisted under all major regimes, but pathway count and strength varied by state. The Wind

→ Humidity fraction was higher in DJF than in JJA, indicating stronger wind-to-moisture dependence under winter conditions. High_humidity and high_cloud regimes had fewer total significant edges than the corresponding normal regimes, but chain scores and mediation reductions were relatively higher, suggesting that humid or cloudy backgrounds may strengthen pathway intensity.

The high-state samples were substantially smaller than their normal-state counterparts, so we repeated the comparison after matched-size resampling. High humidity contained 13,734 samples versus 41,200 normal-humidity samples; high cloud contained 13,734 samples versus 54,934 normal-cloud samples. Across 20 matched repetitions, the mean Jaccard distances were 0.119 and 0.111, respectively, compared with original distances of 0.106 and 0.087. The high-versus-matched-normal difference in total unique edge count was close to zero for humidity states and 15.5 edges for cloud states. Thus, unequal sample size explains part of the raw edge-count contrast but does not remove the state-dependent graph reorganization.

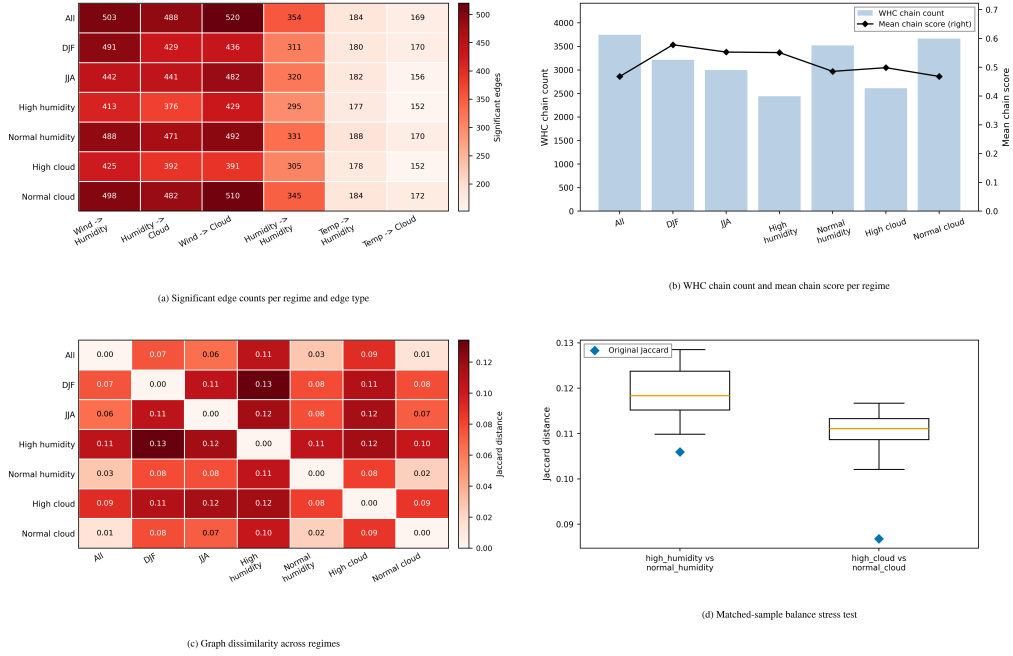


Figure 5: Regime dependence of the WHC pathway: chain counts and scores across seasonal, humidity-state and cloud-state regimes.

Table 4: Regime graph differences measured by Jaccard distance.

Comparison	Jaccard distance	Union edges	Intersection edges
DJF vs JJA	0.107	851	760
high_humidity vs normal_humidity	0.106	840	751
high_cloud vs normal_cloud	0.087	841	768
all vs DJF	0.075	856	792
all vs JJA	0.065	851	796

Table 5: Matched-sample sensitivity for high-state versus normal-state graphs.

Comparison	High samples	Normal samples	Original Jaccard	Matched Jaccard mean	Matched Jaccard SD	Mean edge-count difference
high humidity vs normal humidity	13,734	41,200	0.106	0.119	0.006	-0.75
high cloud vs normal cloud	13,734	54,934	0.087	0.111	0.007	15.50

4.4. Null-model and directionality control experiments

Null-model analysis showed that the core pathway could not be explained by random candidate edges or variable-type composition alone. The significant ratio of the physical candidate graph was 0.792, higher than fully random edges (0.513) and variable-preserved random edges (0.458). Under the variable-preserved random control, Wind $\rightarrow$ Humidity was enriched by 1.77-fold, Humidity $\rightarrow$ Cloud Cover by 1.68-fold, and Wind $\rightarrow$ Cloud Cover by 1.93-fold. The distance-matched random graph still had a high significant ratio of 0.743, indicating that spatial-neighbour correlation is strong. Thus, the appropriate interpretation is not that the result is independent of spatial correlation, but that physically motivated pathways show additional enrichment within a strongly spatially correlated background.

Directionality controls showed that real temporal alignment was essential for the main pathway, but they did not establish a uniquely oriented causal graph. The forward Wind $\rightarrow$ Humidity significant ratio was 0.781, higher than the reverse Humidity $\rightarrow$ Wind ratio of 0.639. After time shuffling, Wind $\rightarrow$ Humidity and Wind $\rightarrow$ Cloud Cover were reduced to zero significant edges, and Humidity $\rightarrow$ Cloud Cover was almost eliminated. Circular shifting reduced the significant ratios of the three main pathways to approximately 0.36–0.38. However, Cloud $\rightarrow$ Humidity and future-to-past Humidity $\rightarrow$ Cloud Cover remained strong. These controls therefore reject explanations based only on arbitrary time alignment, while also indicating bidirectional coupling, persistence or unresolved common forcing in the humidity–cloud subsystem.

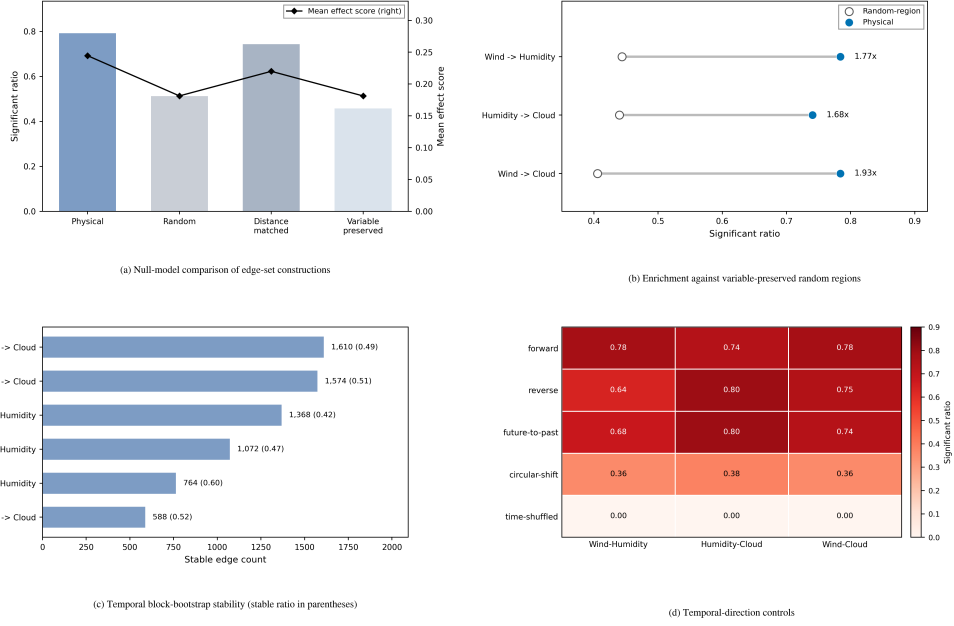


Figure 6: Null-model and directionality controls: physical candidates versus random controls, and pathway behaviour under reversed, shuffled and circularly shifted time.

Table 6: Null-model controls.

Edge set	Tested edges	Significant edges	Significant ratio	Mean absolute coefficient	Mean effect score
physical_candidates	18,018	14,274	0.792	0.0228	0.2446
random_edges	18,018	9,235	0.513	0.0186	0.1814
distance_matched_random	18,018	13,392	0.743	0.0208	0.2200
variable_preserved_random	18,018	8,244	0.458	0.0189	0.1813

Table 7: Enrichment under variable-preserved random controls.

Edge type	True significant ratio	Random-region significant ratio	Enrichment
Wind → Humidity	0.784	0.444	1.77
Humidity → Cloud Cover	0.741	0.440	1.68
Wind → Cloud Cover	0.784	0.406	1.93

Table 8: Directionality and pseudo-time controls.

Test	Edge type	Tested edges	Significant edges	Significant ratio	Mean effect score
forward	Wind $\rightarrow$ Humidity	4,158	3,247	0.781	0.207
reverse	Humidity $\rightarrow$ Wind	4,158	2,655	0.639	0.123
time_shuffled	Wind $\rightarrow$ Humidity	4,158	0	0.000	0.000
circular_shift	Wind $\rightarrow$ Humidity	4,158	1,498	0.360	0.115
forward	Humidity $\rightarrow$ Cloud Cover	4,158	3,076	0.740	0.284
reverse	Cloud $\rightarrow$ Humidity	4,158	3,332	0.801	0.189
time_shuffled	Humidity $\rightarrow$ Cloud Cover	4,158	1	0.000	0.063
circular_shift	Humidity $\rightarrow$ Cloud Cover	4,158	1,585	0.381	0.120
forward	Wind $\rightarrow$ Cloud Cover	4,158	3,258	0.784	0.242
time_shuffled	Wind $\rightarrow$ Cloud Cover	4,158	0	0.000	0.000

4.5. Bootstrap stability, historical splits and robustness tests

Block bootstrap showed that the core edges were reproducible across long-term samples. Wind $\rightarrow$ Humidity had 3,260 significant edges, of which 1,368 met the stability criterion, with mean stability of 0.898. Humidity $\rightarrow$ Cloud Cover had 3,079 significant edges, of which 1,574 were stable, with mean stability of 0.887. Wind $\rightarrow$ Cloud Cover had 3,260 significant edges, of which 1,610 were stable, with mean stability of 0.890.

Historical-period analysis further showed that WH, HC, WC and WHC chains were retained across 1979-1989, 1990-1999, 2000-2009, 2010-2018 and 2019-2025. The overlap between each period graph and the full-period graph remained between 0.883 and 0.910.

Table 9: Bootstrap stability.

Edge type	Significant edges	Stable edges	Stable ratio	Mean stability
Wind $\rightarrow$ Humidity	3,260	1,368	0.420	0.898
Humidity $\rightarrow$ Cloud Cover	3,079	1,574	0.511	0.887
Wind $\rightarrow$ Cloud Cover	3,260	1,610	0.494	0.890
Humidity $\rightarrow$ Humidity	2,261	1,072	0.474	0.910
Temperature $\rightarrow$ Humidity	1,273	764	0.600	0.938
Temperature $\rightarrow$ Cloud Cover	1,141	588	0.515	0.895

Table 10: Historical-period stability.

Period	Time range	Significant ratio	WH edges	HC edges	WC edges	WHC chains	Overlap with full graph
P1	1979-1989	0.734	428	407	421	2,700	0.910
P2	1990-1999	0.714	428	391	411	2,647	0.901
P3	2000-2009	0.724	424	390	427	2,650	0.910
P4	2010-2018	0.715	406	388	418	2,531	0.906
P5	2019-2025	0.678	390	365	386	2,261	0.883

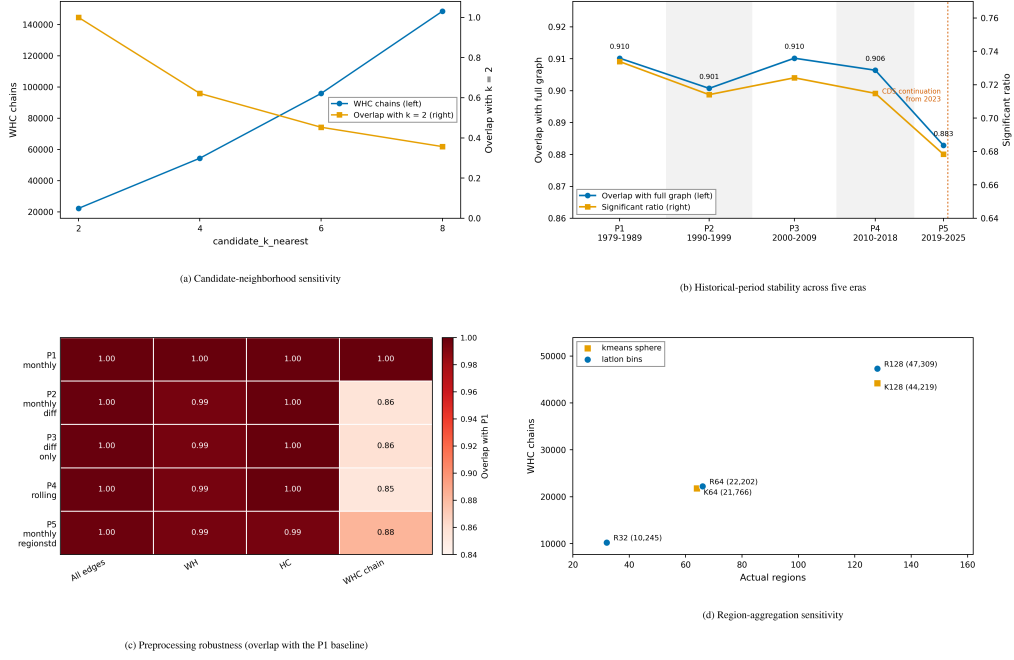


Figure 7: Robustness of the recovered structure under bootstrap, candidate- k , preprocessing and region-aggregation perturbations.

Table 11: Candidate-k robustness.

k	Candidate edges	Tested edges	Significant ratio	Unique significant edges	WH edges	HC edges	WHC chains	Recovered k=2 edges
2	858	18,018	0.792	857	3,260	3,079	22,202	857
4	1,386	29,106	0.746	1,379	4,956	4,926	54,351	857
6	1,914	40,194	0.697	1,892	6,400	6,655	95,871	857
8	2,442	51,282	0.672	2,403	7,804	8,376	148,573	857

Table 12: Preprocessing robustness.

Setting	Significant ratio	WH edges	HC edges	WC edges	WHC chains	WHC chain overlap
P1_monthly	0.792	3,260	3,079	3,260	22,202	1.000
P2_monthly_diff	0.874	3,535	3,630	3,444	27,922	0.856
P3_diff_only	0.878	3,546	3,676	3,442	28,468	0.858
P4_rolling	0.852	3,517	3,427	3,415	26,472	0.854
P5_monthly_regionstd	0.784	3,180	3,141	3,168	22,131	0.883

Table 13: Region-aggregation robustness.

Setting	Method	Actual regions	Significant ratio	WH edges	HC edges	WC edges	WHC chains	Midlatitude strongest
R32	latlon_bins	32	0.785	1,491	1,498	1,573	10,245	no
R64	latlon_bins	66	0.792	3,260	3,079	3,260	22,202	yes
R128	latlon_bins	128	0.822	6,460	6,488	6,358	47,309	yes
K64	kmeans_sphere	64	0.791	3,035	3,161	3,159	21,766	yes
K128	kmeans_sphere	128	0.804	6,075	6,474	6,359	44,219	yes

4.6. Latitude-band, long-lag and humidity-mediation analyses

Latitude-band analysis provided one of the strongest physically interpretable signals. Midlatitude regions had 9,612 WHC chains and 3,628 stable WHC chains, with a mean chain score of 0.642. Tropical regions had 6,017 WHC chains, 1,201 stable WHC chains and a mean chain score of 0.451; polar regions had 6,573, 888 and 0.375, respectively. This midlatitude enhancement is consistent with synoptic systems, frontal activity and storm-track moisture transport-cloud response mechanisms.

The long-lag experiment extended `max_lag` to 12 time steps, corresponding to 72 hours. The peak lag of Wind $\rightarrow$ Humidity, Humidity $\rightarrow$ Cloud Cover and Wind $\rightarrow$ Cloud Cover remained concentrated at lag 1, but the mean lag was approximately 5.9-6.3 time steps, and the lag 4-8 range, corresponding to 24-48 hours, accounted for a large fraction. Humidity mediation tests showed that after controlling for Humidity, Wind $\rightarrow$ Cloud Cover coefficients decreased overall but only modestly. The mean reduction was 0.037 in the all regime and approximately 0.104 and 0.109 in high_humidity and high_cloud regimes, respectively.

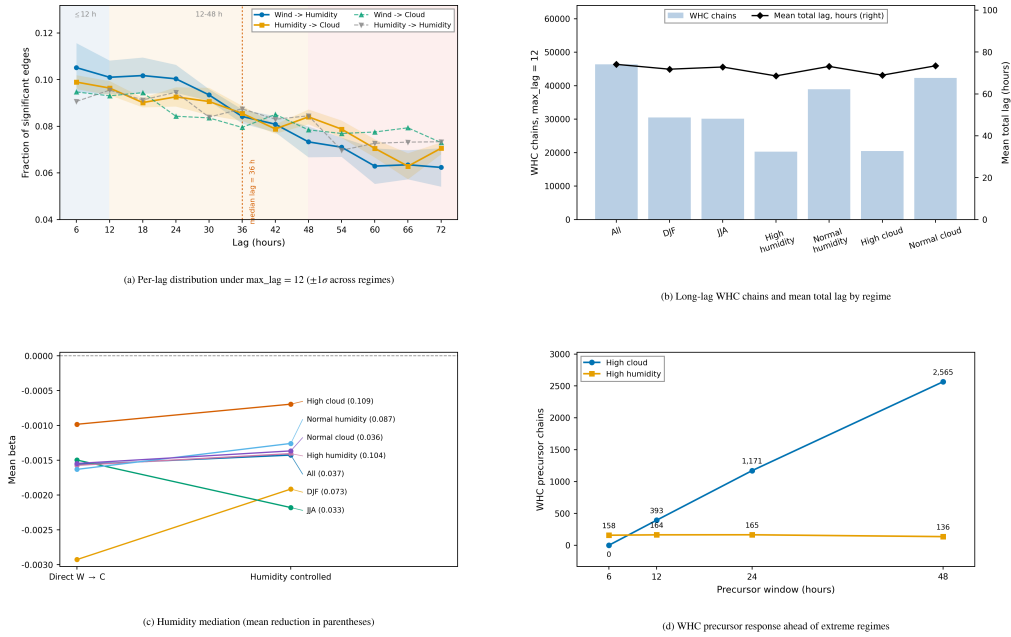


Figure 8: Mechanism boundaries of the pathway: latitude-band structure, long-lag behaviour and humidity mediation of Wind $\rightarrow$ Cloud Cover.

Table 14: Latitude-band WHC chain statistics.

Latitude band	WHC chains	Stable WHC chains	Mean total lag	Mean chain score
tropical	6,017	1,201	3.985	0.451
midlatitude	9,612	3,628	3.983	0.642
polar	6,573	888	3.941	0.375

Table 15: Long-lag experiment with max_lag=12.

Edge type	Peak lag	Mean lag	Median lag	Lag 1 fraction	Lag 2-3 fraction	Lag 4-8 fraction	Lag 9-12 fraction
Wind → Humidity	1	5.89	6.0	0.103	0.200	0.433	0.263
Humidity → Cloud Cover	1	6.09	6.0	0.099	0.186	0.431	0.284
Wind → Cloud Cover	1	6.25	6.0	0.094	0.187	0.412	0.307
Humidity → Humidity	2	6.19	6.0	0.090	0.186	0.433	0.291

4.7. Multivariable adjustment and effect-size analysis

Controlling the complete recent multivariate state of the target region reduced the number of all-regime significant edge-lag hypotheses from 2,218 to 2,163, but retained 1,920 of the original discoveries. Wind → Humidity, Humidity → Cloud Cover and Wind → Cloud Cover retained 439/503, 407/488 and 428/520 original significant hypotheses, corresponding to recovery fractions of 0.873, 0.834 and 0.823. The adjusted graph still contained 3,624 WHC chains, compared with 3,746 in the principal model. The core pathway is therefore not removed by controlling recent temperature, humidity, wind and cloud history in the target region.

The median effects across all significant hypotheses were statistically clear but individually small. Under multivariable adjustment, median absolute standardized coefficients were 0.0138, 0.0207 and 0.0176 for WH, HC and WC, while median partial R^2 values were 0.00117, 0.00090 and 0.00084. This distinguishes the widespread graph-level recurrence of the pathway from the magnitude of any single edge. In contrast, the ten strongest stable edges per pathway had median absolute coefficients of 0.195–0.328 and median partial R^2 of 0.035–0.042. All 30 of these selected edges had year-block bootstrap 95% intervals excluding zero, with median sign consistency of 1.0 across the

Table 16: Humidity mediation test for Wind → Cloud Cover.

Regime	WC edges	Mean direct beta	Mean beta after controlling Humidity	Mean reduction	Reduction > 30%	Reduction > 50%
all	520	-0.00156	-0.00143	0.0369	0.206	0.117
DJF	436	-0.00293	-0.00191	0.0729	0.211	0.108
JJA	482	-0.00150	-0.00218	0.0330	0.199	0.100
high_humidity	429	-0.00157	-0.00140	0.1045	0.182	0.091
normal_humidity	492	-0.00163	-0.00126	0.0873	0.215	0.118
high_cloud	391	-0.00098	-0.00069	0.1090	0.192	0.079
normal_cloud	510	-0.00155	-0.00137	0.0358	0.198	0.102

47 annual blocks.

Table 17: All-regime multivariable-control and effect-size summary.

Edge type	Principal significant	Adjusted significant	Original retained	Recovery	Median $ \beta $	Median partial R^2	Top-edge block CI exclusion
Wind $\rightarrow$ Humidity	503	490	439	0.873	0.0138	0.00117	10/10
Humidity $\rightarrow$ Cloud Cover	488	483	407	0.834	0.0207	0.00090	10/10
Wind $\rightarrow$ Cloud Cover	520	487	428	0.823	0.0176	0.00084	10/10

4.8. PCMCI confirmation on the recent-period sample

PCMCI was run for 60 top all-regime stable Granger-style edges using the 2019–2025 data segment at 6-hour resolution. After conditional-independence testing and Benjamini–Hochberg correction, 18 of 20 Wind $\rightarrow$ Humidity edges, 16 of 20 Humidity $\rightarrow$ Cloud Cover edges and 16 of 20 Wind $\rightarrow$ Cloud Cover edges remained significant. Mean absolute PCMCI partial correlations were 0.0867, 0.0426 and 0.0815, respectively. This confirmation does not remove the observational assumptions of PCMCI, but it shows that most strongest stable edges survive a separate conditional-independence discovery procedure in the recent historical period.

Table 18: PCMCI validation of top stable all-regime edges over 2019–2025.

Edge type	Granger top edges	PCMCI confirmed	Confirmed ratio	Mean absolute partial correlation	Status
Wind $\rightarrow$ Humidity	20	18	0.900	0.0867	confirmed
Humidity $\rightarrow$ Cloud Cover	20	16	0.800	0.0426	confirmed
Wind $\rightarrow$ Cloud Cover	20	16	0.800	0.0815	confirmed

5. Discussion and Limitations

5.1. Interpretation and physical implications

The results support a cautious but clear interpretation: in 6-hourly global reanalysis data from 1979 to 2025, Wind $\rightarrow$ Humidity $\rightarrow$ Cloud Cover appears as a stable, non-random and reproducible lagged directed dependency pathway across regimes and historical periods. The pathway is enriched in the physical candidate graph, remains approximately 1.7-fold enriched under variable-preserved random controls, survives block bootstrap and historical-period splits, and is robust to preprocessing and region-aggregation choices.

The most important mechanistic signal is the midlatitude enhancement. Midlatitude regions have more chains, more stable chains and higher mean chain scores than tropical or polar regions. This pattern is consistent with

synoptic-scale moisture advection, frontal lifting and cloud response. Causal WeatherGraph therefore provides more than a correlation graph: it links statistical dependency, spatial organization and physical interpretation in a regional evidence framework.

The regime-aware results indicate that atmospheric dependency graphs should not be treated as static structures. In DJF, the Wind $\rightarrow$ Humidity fraction and chain score are relatively high, suggesting that wind-to-moisture lagged dependence is stronger under winter conditions. Under high-humidity and high-cloud states, chain counts are not necessarily largest, but chain scores and mediation reductions tend to be stronger. Regime changes therefore affect not only the number of edges but also pathway intensity and interpretation.

The multivariable and effect-size analyses sharpen this interpretation. Most all-regime discoveries and nearly all core pathway structure remain after controlling the recent four-variable state of the target region, and PCMCi confirms 80–90% of the selected top stable edges in 2019–2025. At the same time, median partial R^2 values across all significant core edges are below 0.002. The result is therefore a distributed, recurrent graph-level signal rather than a claim that every individual atmospheric edge has a large effect. The strongest stable edges form a smaller subset with materially larger standardized coefficients, partial R^2 and year-block consistency.

Several boundaries are essential. First, all results are based on observational reanalysis data and cannot replace intervention experiments. A significant edge means that a lagged source term statistically increases explanatory power for a target, not that intervening on wind would necessarily change humidity and then cloud cover. Second, atmospheric variables have strong autocorrelation, common weather-system forcing and spatial-neighbour coupling. This is especially important for humidity and cloud cover, where reverse and future-to-past controls remain strong. Third, the high significant ratio of distance-matched random edges shows that part of the dependency structure arises from spatial-neighbour background. Fourth, humidity mediation explains only part of Wind $\rightarrow$ Cloud Cover, so the WHC pathway should be described as an important channel, not as the only channel.

These boundaries strengthen rather than weaken the contribution. Causal WeatherGraph offers a reproducible workflow: restrict the search space with physically interpretable candidate edges, compare graph structures under atmospheric regimes, and use layered controls and robustness tests to check whether a pathway still holds. For atmospheric research that seeks mecha-

nistic interpretation in addition to forecast skill, this framework provides a scalable template for analysing variable propagation.

From an information-science perspective, the same design pattern — domain-constrained candidate generation to control the hypothesis space, regime-conditioned estimation to expose state-dependent structure, and a layered chain of matched nulls, falsification tests and confirmatory re-estimation to control the quality of discovered structure — is not specific to the atmosphere. It applies wherever directed lagged dependencies must be extracted from large, autocorrelated and partially confounded spatio-temporal records, including traffic and mobility networks, energy systems, hydrology, air quality and epidemiological surveillance.

5.2. Limitations and future directions

Our study has four main limitations. First, it uses observational reanalysis data and does not provide interventional causal proof. Granger-style and PCMCI-significant edges remain conditional observational dependencies. Second, although monthly climatology was removed and multiple preprocessing robustness tests were performed, strong autocorrelation, common weather systems and spatial smoothing may still affect edge significance. Third, regional aggregation improves interpretability and computational feasibility, but it can mix local processes within each region. Fourth, confirmatory PCMCI was restricted to selected stable edges and the 2019–2025 segment, and both the principal estimator and PCMCI partial-correlation test are linear.

Future work can strengthen the framework in four directions. First, full-period PCMCI+, nonlinear conditional independence tests or kernel-based causal discovery can reassess the strongest stable edges. Second, additional physical stratifications, including land-sea contrast, topography, vertical levels, storm-track domains and large-scale circulation indices, can be incorporated. Third, the current linear Granger-style OLS backbone can be extended to nonlinear or graph-neural causal models. Fourth, the existing extreme-event precursor module can be extended to precipitation outcomes and evaluated prospectively.

6. Conclusion

We introduced Causal WeatherGraph for identifying regime-aware regional lagged directed dependency graphs from long-term global reanalysis

data. The method aggregates gridded atmospheric variables into regional time series, constructs physically motivated candidate edges, and identifies significant lagged dependencies through Granger-style OLS and FDR correction. Around the Wind $\rightarrow$ Humidity $\rightarrow$ Cloud Cover pathway, we build an evidence chain involving null models, directionality controls, bootstrap stability, latitude-band analysis, historical-period splits, candidate-neighbourhood expansion, preprocessing robustness and region-aggregation robustness.

The main finding is that Wind $\rightarrow$ Humidity and Humidity $\rightarrow$ Cloud Cover appear consistently across all major regimes. In the all regime, 3,746 WHC chains were detected, with a mean total lag of approximately 24 hours; 3,624 chains remained after target-region multivariable adjustment. The physical candidate graph is enriched relative to random controls, depends on real temporal alignment, is reproducible across time blocks and historical periods, and is strongest in midlatitude regions. PCMCI confirms most selected top stable edges, while effect-size analysis shows that the typical individual edge is small. The appropriate conclusion is therefore not that the study proves an interventional chain, but that it identifies a distributed, physically consistent and repeatedly validated Wind $\rightarrow$ Humidity $\rightarrow$ Cloud Cover lagged dependency pathway. Beyond this case study, Causal WeatherGraph offers a general and scalable template for statistically controlled discovery of directed dependency structure in large spatio-temporal information systems, and its constraint-based search-space reduction, regime-conditioned estimation and layered validation chain can be instantiated in other domains with domain-appropriate candidate constraints.

Data and code availability

The analysis code, configurations and derived summary tables are available at <https://github.com/gengx-prog/causal-weathergraph>. ERA5 data are available from the Copernicus Climate Data Store, and the preprocessed benchmark-era segment is available from the public WeatherBench 2 archive.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have influenced the work reported in this paper, and is not a member of the editorial board of this journal.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the author used Claude (Anthropic) for language editing to improve readability, organization and journal fit. The author reviewed and edited all generated content and takes full responsibility for the final manuscript.

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:



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